PROCUREMENT / PROGRAMME DEFINITION

Prototype P0-V1 Parts, Cost & R&D;

VTX-P0-V1 / P0-V1 / ISSUE A / 30 OCT 2026

ENGINEERING CONCEPT / NOT A RELEASED DESIGN: geometry is an internally consistent concept baseline; component masses, aerodynamic coefficients, engine-specific fuel consumption, crashworthiness and flight-control performance are not substantiated by qualification tests.

One-off, two-place, gimballed spherical-cabin VTOL with a central CFRP ring and four ducted rotors. This document defines an initial procurement and development investment envelope, not a production quotation.

01 / Executive cost position

COST GROUP	LOW PLANNING CASE	HIGH PLANNING CASE
P0 component procurement / fabrication	GBP 1,070,000	GBP 3,110,000
Non-recurring development & validation	GBP 3,970,000	GBP 12,050,000
Subtotal	GBP 5,040,000	GBP 15,160,000
Explicit programme contingency: 25%	GBP 1,260,000	GBP 3,790,000
TOTAL DEVELOPMENT ENVELOPE	GBP 6,300,000	GBP 18,950,000

R&D; budget is separate from the listed completed-P0 parts budget. Test articles, coupons, sacrificial rigs and prototype rework may appear within R&D.; All figures are illustrative GBP 2026 planning allowances; no supplier quotations, VAT allocation or contracts verified. Unlimited R&D; budget means the upper range is not a spending cap.

Configuration and controlled cost basis

METRIC	VALUE	COST / MASS BASIS
Empty mass target	671 kg	11 subsystem allocations, P01-P11
Test loaded mass	901 kg	671 + 170 kg occupants + 60 kg Jet A-1
Nominal turbine reference	M250-C30P	650 shp takeoff; supplier/install fit review outstanding
Fan group	4 x 2.257 m	16.0 m² disk area, composite ducts and blades
Reference fuel price	GBP 1.45/L	Brighton City Airport gross listed, 5 Sep 2026 [S10]

02 / Component procurement ledger

ID	SUBSYSTEM	KG	LOW GBP k	HIGH GBP k
P01	Turboshaft installation	132	220	650
P02	Four ducted fan modules	119	120	320
P03	Four-way transmission	72	130	350
P04	Spherical cabin module	88	120	400
P05	Two-axis cabin gimbal	35	80	230
P06	Main structure	75	120	350
P07	Fuel installation	29	35	100
P08	Flight-control avionics	23	110	280
P09	Two seat/restraint systems	25	25	90
P10	Landing gear	24	30	110
P11	Electrical and cooling	49	80	230
-	TOTAL TARGET EMPTY MASS	671	1,070	3,110

P0-V1 masses are held fixed across the technical PDF and blueprint slides. Structural reserve/mass-growth allowance is NOT silently added here; mass-growth scenarios must be reported as explicit deltas.

Cost-estimate classification

Indicative order-of-magnitude engineering allowances for one bespoke full-scale prototype. Engine cost depends critically on serviceable status, overhaul condition and availability. Duct/manufacturing prices depend on composite layup, tool count, quality records and iterative test rejects. Do not treat these numbers as supplier offers.

03 / What is included in each subsystem

P01 Turboshaft installation / 132 kg

M250-C30P class; mounts, intake, exhaust and ancillaries. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P02 Four ducted fan modules / 119 kg

CFRP blades, short ducts, stators, independently actuated collective-pitch hubs, containment. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P03 Four-way transmission / 72 kg

Common-speed reduction, shafts, couplings, gearboxes and bearings. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P04 Spherical cabin module / 88 kg

CFRP sandwich shell; laminated glazing; internal liner. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P05 Two-axis cabin gimbal / 35 kg

Gimbal rings, bearings, actuation and sensing. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P06 Main structure / 75 kg

40 kg ring allocation plus four stiffened radial supports. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P07 Fuel installation / 29 kg

Protected tanks, lines, pumps, vents, instrumentation. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P08 Flight-control avionics / 23 kg

Independent control computer, IMU, sensors, wiring. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P09 Two seat/restraint systems / 25 kg

Load-bearing seats, harnesses and anchors. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P10 Landing gear / 24 kg

Skid/strut arrangement with energy absorbing elements. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

P11 Electrical and cooling / 49 kg

Power buses, actuators, heat exchangers, auxiliaries. The allocation covers a component-level design target; detailed drawings, measured masses, interfaces and supplier selections are pending.

04 / Materials and weight-risk strategy

LOCATION	PRIMARY STUDY MATERIAL	VERIFICATION ISSUE
Cabin structural shell	CFRP skins / aramid honeycomb or PMI foam core	Buckling, face/core debond, impact, fire/smoke, ingress
Cabin transparent regions	Laminated aviation glazing, approx. 35% area target	Pane stack, impact, edges, optical distortion; 88 kg cabin allowance
Interior trim	Flame-resistant PEI/PEEK where justified; PVC only if qualified	Fire/smoke/toxicity; non-load-bearing only
Central ring and supports	Tailored CFRP box structures + metallic inserts	Local compression, torsion, damage tolerance, joint slip
Bearing and gearbox interfaces	Titanium and steels where needed	Bearing pressure, fatigue, temperature, lubrication
Rotor/stator ducts	Composite structure with containment concept	Blade-out load, dynamic distortion and tip-gap clearance

Why weight targets are not yet released

The 75 kg frame allocation uses an illustrative 40 kg central ring plus four internally stiffened arms of approximately 8.76 kg each. Shortening these arms requires the central ring to accept the transferred loads. Local panel buckling may invalidate the selected 3.5 mm skins without verified transverse stiffness and webs. The 88 kg cabin includes a speculative 14 kg increment over the earlier monolithic-glass concept to allow laminated glazing.

05 / Non-recurring R&D; programme

ID	WORK PACKAGE	LOW GBP k	HIGH GBP k
R01	Architecture, requirements, configuration control	220	550
R02	Composite design, coupon tests, joints and nonlinear FEA	300	900
R03	Aerodynamic CFD, wind tunnel and airframe interference	350	1,100
R04	Rotor/stator, variable-pitch hubs, tooling and dynamometer	450	1,400
R05	Transmission, lubrication, turbine integration test rigs	450	1,400
R06	Gimbal / flight-controls software, HIL and fault injection	450	1,200
R07	Drop, landing, impact, fatigue and structural proof tests	250	800
R08	Manufacturing tooling, quality assurance and assembly	450	1,300
R09	Tethered / uncrewed flight-test and instrumented campaign	400	1,200
R10	Test site, telemetry, rigs and data instrumentation	350	1,300
R11	Independent assurance, test operations and insurance	300	900
-	R&D SUBTOTAL	3,970	12,050

Packages include engineering labour, models, material and destructive test specimens, commissioned tests, revisions and independent review. They do not include the full completed P0 prototype procurement subtotal already listed in section 02.

06 / Development gates and evidence requirements

GATE	OBJECTIVE	MINIMUM DOCUMENTED EVIDENCE
G0	Freeze dimensions, requirements and failure analysis	Signed interface control drawing; load cases; project configuration ledger
G1	Converged structural sizing	Laminate schedules; modal + buckling FEA; coupon/element evidence; measured mass
G2	Rotor and turbine bench substantiation	Thrust/torque maps, energy efficiency, nacelle crosswind, SFC and controls
G3	Gimbal and avionics hardware-in-loop	Sensor-failure test, asymmetric loads, saturation and power interruption
G4	Full-aircraft static/ground test	Proof load, drop/load paths, blade containment assessment, resonance
G5	Tethered uncrewed powered hover	Instrumented margins, fault detection, power oscillation and thermal limits
G6	Uncrewed restricted-range manoeuvre programme	500 m baseline and 200 m conditional turn; repeatable flight logs
G7	Post-test design release decision	Revised actual masses, measured power map and unresolved hazard closure

07 / Principal engineering uncertainties

ID	RISK	SPECIFIC IMPACT
U01	Four-thrust-channel control loss	The simplified allocation loses a degree of independent control after one rotor fails; backup torque/authority not demonstrated.
U02	Common turbine failure	All four rotor branches share one prime mover; no power-out continued-hover capability in P0.
U03	Drive torque & dynamics	Gear reduction, shaft critical speeds and torsional vibration under cyclic rotor loading unverified.
U04	Duct aero / structural distortion	Blade tip clearance, edgewise-flow losses, rotor wake interference and containment not yet tested.
U05	Cabin impact and gimbal integrity	Laminated glazing and composite panel impact/load allowables not substantiated.
U06	Mass budget / CG	No solid CAD or weighed subassembly; fore-aft fuel/crew balance and mass moments of inertia unknown.
U07	SFC & power integration	Reference 0.4 kg/kWh independent of engine power and altitude is unvalidated; rated power not constant.
U08	Test configuration maturity	Drawings are parametric concept views, not fabrication or toleranced machining drawings.

Development spending priorities

Do not first optimise only cost per kilogram. Priority 1 is a testable rotor/thrust/engine envelope; priority 2 a verified stiff structural load path; priority 3 control authority with asymmetric loads and credible propulsion-failure response; priority 4 aerodynamic optimisation of duct lip, blade and stator; priority 5 component-level production design.

Budget sensitivity

Changing fuel from GBP 1.45/L to GBP 1.82/L (both published UK airfield examples, [S10-S11]) alters P0-V1 calculated 27 km mission fuel cost from approximately GBP 45.58 to GBP 57.22 at the nominal assumed engine efficiency. That fuel variation is small compared with airframe development costs.

08 / Research and provenance

S01 Rolls-Royce: M250/RR300 FIRST Network Directory (2024); C30P 650 shp takeoff, nominal 245 lb dry, published SFC table. OPEN SOURCE

S02 Bell / 206L-4: Engine takeoff and maximum-continuous engine ratings vs installed transmission limitations (2005 product specification). OPEN SOURCE

S03 Hine et al., 2025: Design and Analysis of a Liftfan for eVTOL Aircraft. Experimental rotor/stator optimisation and nacelle forward-flight losses. DOI:10.1115/1.4067673. OPEN SOURCE

S04 Hu et al., 2021: Hovering efficiency optimization of the ducted propeller with weight penalty taken into account. DOI:10.1016/j.ast.2021.106937. OPEN SOURCE

S05 Akturk & Camci, 2014: Tip-clearance treatment experiments in ducted VTOL fans. DOI:10.1115/1.4023469. OPEN SOURCE

S06 Critchfield & Ning, 2026: Mission-focused multidisciplinary eVTOL propulsion optimisation. DOI:10.2514/1.C038445. OPEN SOURCE

S07 Li, Chen & Jia, 2023: Experimental investigation on hover performance of a ducted coaxial-rotor UAV; rotor location and clearance influence. OPEN SOURCE

S08 Liu et al., 2024: Impact response and crashworthy design of composite fuselage structures: An overview. OPEN SOURCE

S09 Muniraj & Sreehari, 2021: Experimental damage evaluation of honeycomb sandwich with composite face sheets. OPEN SOURCE

S10 Brighton City Airport: Jet A-1 gross indicative pump price GBP 1.45/L from 5 September 2026. OPEN SOURCE

S11 Shobdon Airfield: Jet A-1 quoted GBP 1.82/L nonmember, 2 October 2026; regional price variation. OPEN SOURCE

Research screening conducted with Wolfram calculations, SciSpace and Consensus; retrieved papers support trade-study assumptions but do not qualify the specific aircraft. Elicit access unavailable in this chat. Source dates and contents should be rechecked during formal design control.